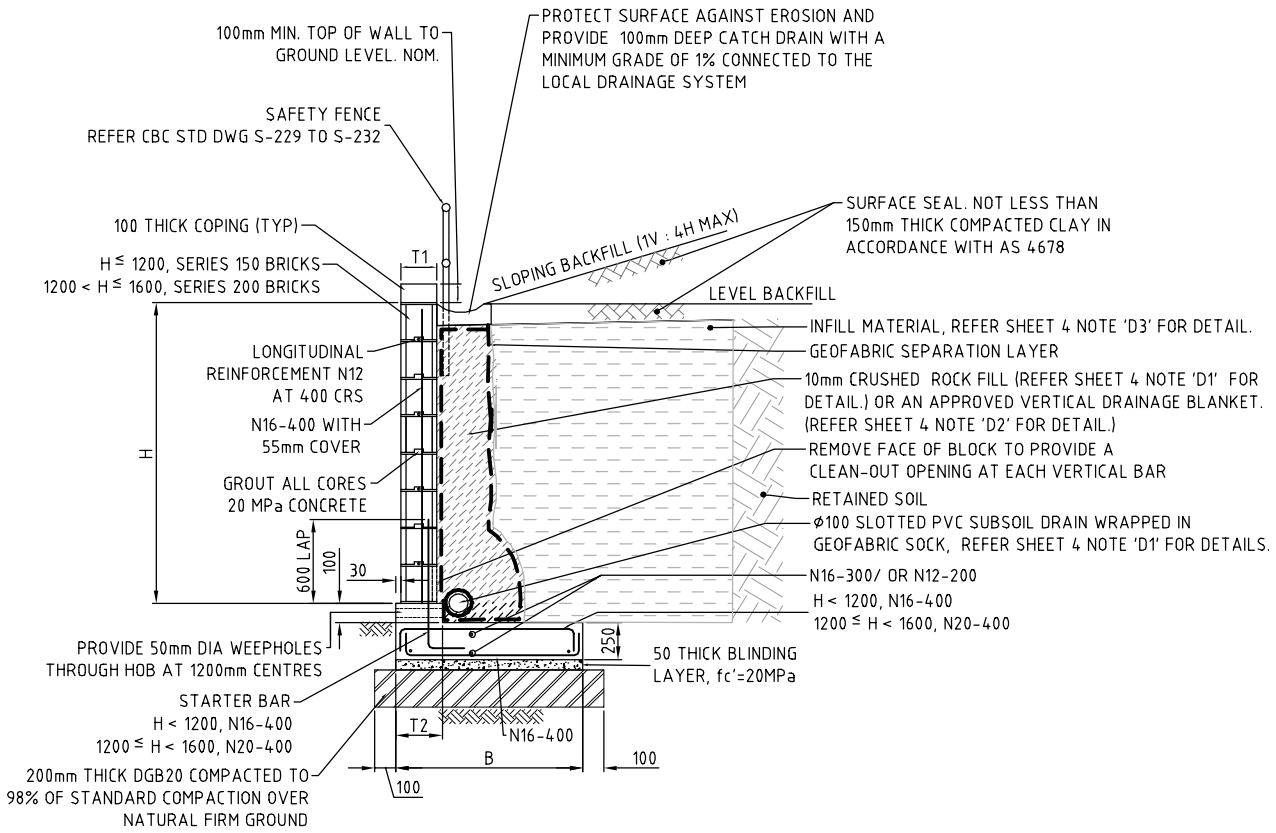


CONCRETE BLOCK RETAINING WALL TYPE 1, RW-CB1
MAXIMUM HEIGHT 1600mm
SCALE: NTS

BACKFILL				
H	T1	T2	B, LEVEL BACKFILL	B, SLOPING BACKFILL *
600	140	200	600	600
800	140	200	700	800
1000	140	200	900	1000
1200	190	250	1100	1200
1400	190	250	1200	1400
1600	190	250	1400	1600

*MAXIMUM SLOPE SHALL NOT EXCEED 1 (VERTICAL) : 4 (HORIZONTAL)

WALL DIMENSIONS TABLE FOR TYPE 1

CONCRETE				
ELEMENT	SLUMP (mm)	AGGREGATE (MAX SIZE IN mm)	CEMENT TYPE	f'c
WALL FOOTING	80	20	GP	32MPa
GROUT INFILL	230	10	GP	20MPa

REINFORCEMENT COVER SCHEDULE				
ELEMENT	COVER (mm)			EXPOSURE CLASSIFICATION
	TOP	BOTTOM	SIDES	
FOOTING	60	70	60	B2
WALL	65	N/A	65	B1

CONCRETE & REINFORCEMENT SPECIFICATIONS

NOTES:
1. FOR GENERAL NOTES, REFER SHEET 4, S-220
2. FOR TYPICAL DETAILS, REFER SHEET 3, S-220
3. WALL HEIGHT H > 1600 SHALL BE DESIGNED BY A QUALIFIED STRUCTURAL DESIGNER
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.

DESIGN MUST BE VERIFIED BY A QUALIFIED ENGINEER FOR SITE SPECIFIC CONDITIONS

CANTERBURY BANKSTOWN

CITY OF CANTERBURY BANKSTOWN

STANDARD RETAINING WALL
CONCRETE BLOCK

RW-CB1

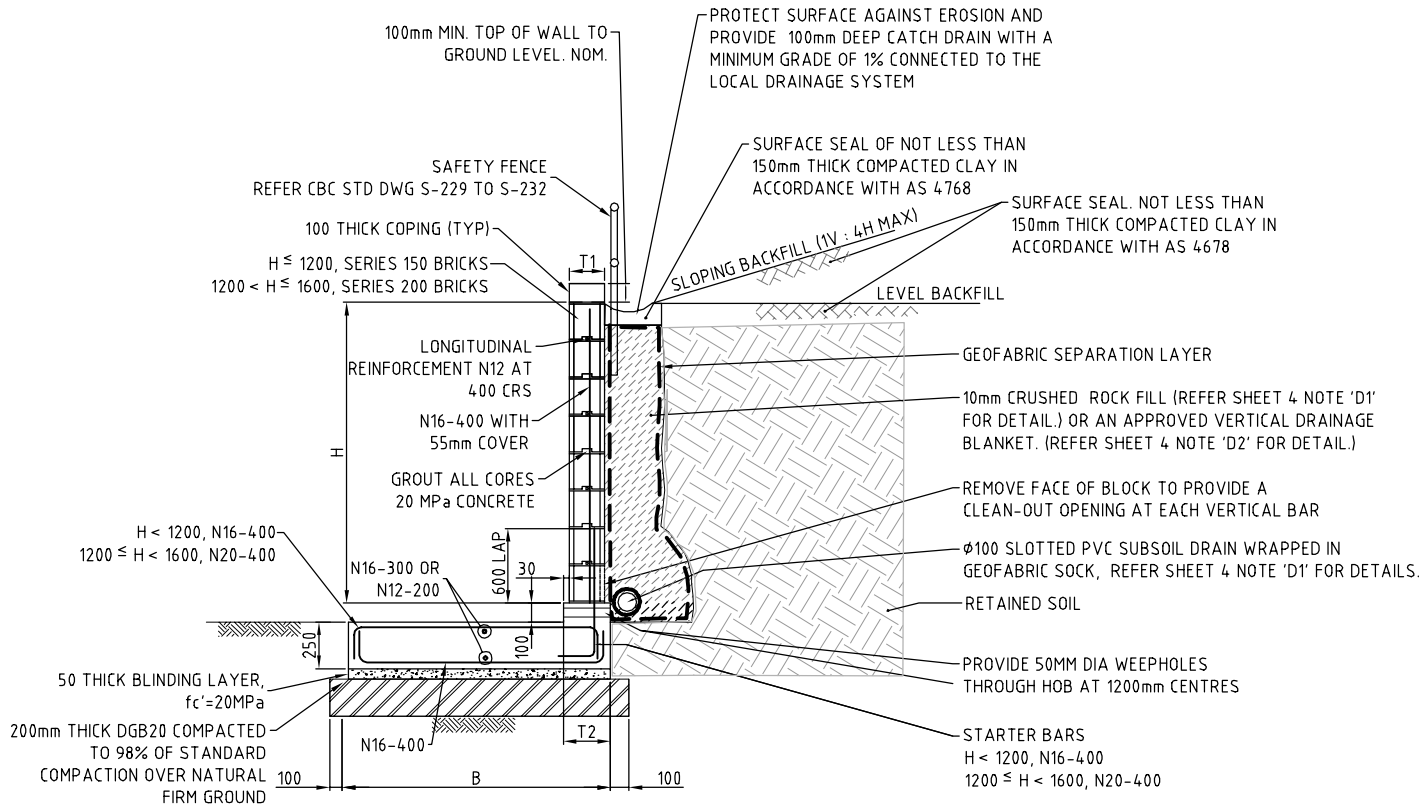
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CONCRETE BLOCK RETAINING WALL TYPE 2, RW-CB2
MAXIMUM HEIGHT 1600mm
SCALE: NTS

BACKFILL				
H	T1	T2	B, LEVEL BACKFILL	B, SLOPING BACKFILL *
600	140	200	600	600
800	140	200	700	800
1000	140	200	900	1000
1200	190	250	1100	1300
1400	190	250	1200	1600
1600	190	250	1500	1900

* MAXIMUM SLOPE SHALL NOT EXCEED 1 (VERTICAL) : 4 (HORIZONTAL)

WALL DIMENSIONS TABLE FOR TYPE 2

CONCRETE				
ELEMENT	SLUMP (mm)	AGGREGATE (MAX SIZE IN mm)	CEMENT TYPE	f'c
WALL FOOTING	80	20	GP	32MPa
GROUT INFILL	230	10	GP	20MPa

REINFORCEMENT COVER SCHEDULE				
ELEMENT	COVER (mm)			EXPOSURE CLASSIFICATION
	TOP	BOTTOM	SIDES	
FOOTING	60	70	60	B2
WALL	65	N/A	65	B1

CONCRETE & REINFORCEMENT SPECIFICATIONS

NOTES:
1. FOR GENERAL NOTES, REFER SHEET 4, S-220
2. FOR TYPICAL DETAILS, REFER SHEET 3, S-220
3. WALL HEIGHT H > 1600 SHALL BE DESIGNED BY A QUALIFIED STRUCTURAL DESIGNER
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.

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CANTERBURY BANKSTOWN

CITY OF CANTERBURY BANKSTOWN

STANDARD RETAINING WALL
CONCRETE BLOCK

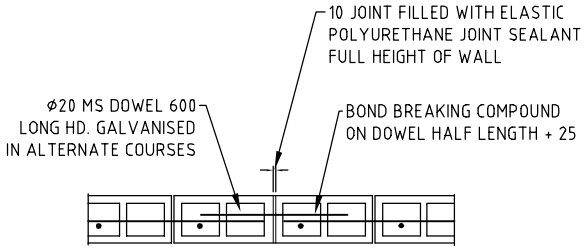
RW-CB2

STD DWG N°
S-220

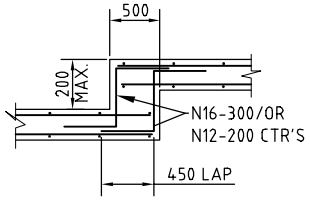
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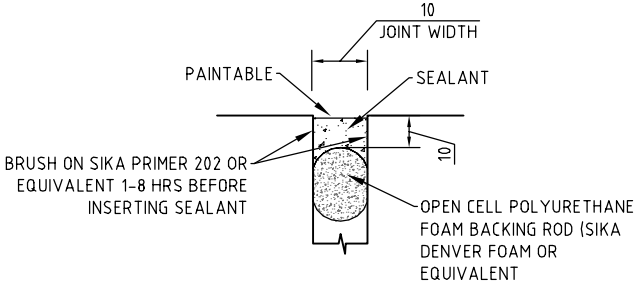
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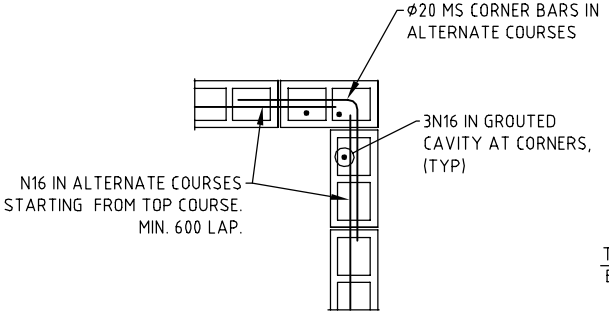
TYPICAL CONTROL JOINT DETAIL
SCALE:NTS



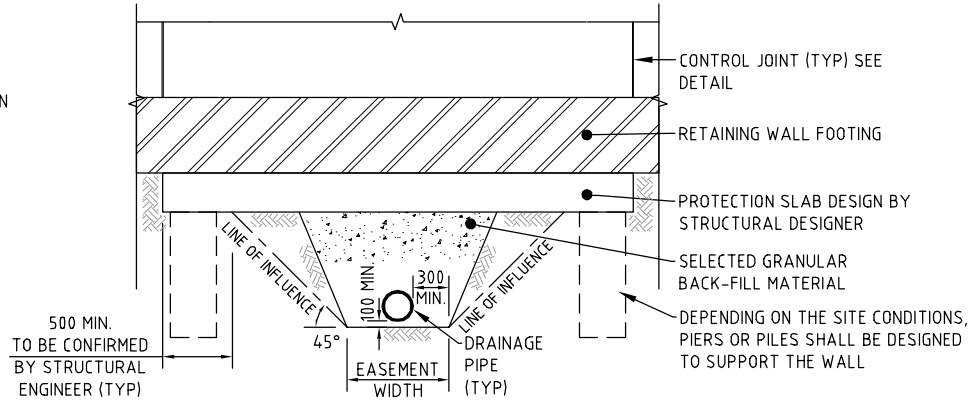
STEPPED FOOTING DETAIL
DETAIL A
SCALE:NTS



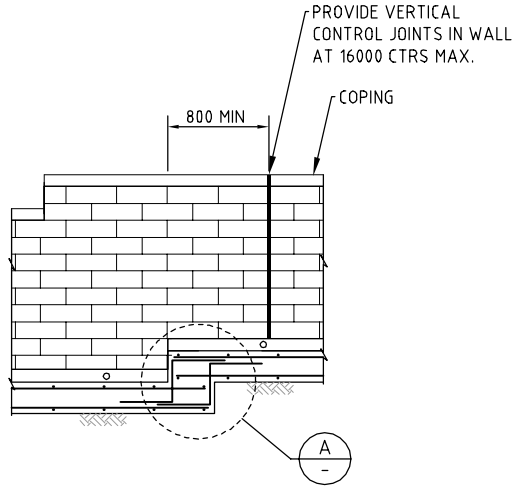
CONTROL JOINT DETAIL
SCALE:NTS



CORNER DETAIL
SCALE:NTS



RETAINING WALL ACROSS DRAINAGE EASEMENT
SCALE:NTS



TYPICAL SECTION - FOOTING STEP
SCALE:NTS

NOTES:
1. FOR GENERAL NOTES, REFER SHEET 4, S-220
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.

DESIGN MUST BE VERIFIED BY A QUALIFIED ENGINEER FOR SITE SPECIFIC CONDITIONS

CANTERBURY BANKSTOWN

CITY OF CANTERBURY BANKSTOWN

STANDARD RETAINING WALL
CONCRETE BLOCK

RW-CB DETAILS

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CONCRETE MASONRY NOTES

M1 ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AUS-SPEC 'MASONARY WALLS' AS3700, AND THE CONCRETE MASONRY ASSOCIATION OF AUSTRALIA'S "REINFORCED CONCRETE MASONRY CANTILEVER RETAINING WALLS - DESIGN AND CONSTRUCTION GUIDE".

M2 CONCRETE BLOCKS TO HAVE A MINIMUM CHARACTERISTIC UNCONFINED COMPRESSIVE STRENGTHS OF f'uc = 15MPa UNO. BLOCK USED IN RETAINING WALLS SHALL BE H BLOCKS (2048) OR DOUBLE U BLOCKS (2091,1591) OR APPROVED EQUIVALENT

M3 ALL CORES ARE TO BE FILLED WITH READY MIX GROUT WITH THE FOLLOWING PROPERTIES.

CEMENT	: 1 PART
HYDRATED LIME	: UPTO 1/10 PART
MORTAR SAND	: 3 PART
10mm AGGREGATE	: UPTO 2 PART

M4 CLEANOUT OPENINGS SHALL BE PROVIDED AT THE BASE OF ALL CORES. ADDITIONAL CLEANOUT OPENINGS ARE REQUIRED AT ALL HORIZONTAL BREAKS IN CONSTRUCTION.

M5 MORTAR FINS PROTRUDING FROM JOINTS SHALL BE REMOVED BEFORE GROUTING CORES.

M6 GROUT SHALL NOT BE POURED INTO CORES FROM ANY HEIGHT GREATER THAN 3m. STOP POUR 50mm BELOW TOP OF BLOCK TO PROVIDE KEY FOR THE FOLLOWING POUR.

M7 RODDING OR OTHER APPROVED MEANS SHALL BE USED TO ENSURE PROPER COMPACTION OF GROUT IN CORES.

M8 TEMPORARY BRACING SHALL BE PROVIDED TO WALLS AS NECESSARY TO MAINTAIN STABILITY DURING CONSTRUCTION.

M9 PROVIDE 50mm DIA WEEP HOLES AT 1200mm CENTRES THROUGH HOB

M10 PROVIDE 10mm VERTICAL CONTROL JOINTS USING ELASTIC JOINT SEALANT AND DOWELS AT MAXIMUM CENTRES OF 16000mm IN ACCORDANCE WITH TYPICAL CONTROL JOINT DETAILS.

DRAINAGE

D1 10mm CRUSHED ROCK FILL BEHIND THE WALL, WRAPPED IN GEOFABRIC.
MINIMUM 300mm THICK AROUND SUBSOIL DRAIN.
SUBSOIL DRAINAGE PIPES TO BE LAID TO A MINIMUM UNIFORM GRADE OF 1%. THE LOW END OF EACH RUN IS TO BE DRAINED THROUGH THE HOB OR DIRECTLY CONNECTED TO A STORM WATER SYSTEM. THE UPPER END IS TO BE BROUGHT TO THE SURFACE AND CAPPED.

D2 PROPRIETARY VERTICAL SHEET DRAINAGE WITH STRIP DRAIN AT BOTTOM OF WALL, INSTALLED IN ACCORDANCE WITH MANUFACTURERS SPECIFICATIONS.

D3 GW(WELL-GRADED GRAVEL) OR SW(WELL-GRADED SAND), PLACED IN LAYERS NOT EXCEEDING 200mm AT A MOISTURE CONTENT WITHIN 2% OF OPTIMUM MOISTURE CONTENT(OMC) TO ACHIEVE 85% STANDARD COMPACTION.

REINFORCED CONCRETE NOTES

RC1 EXPOSURE CLASSIFICATION AND MINIMUM COVER TO REINFORCEMENT SHALL COMPLY WITH THE SCHEDULES SHOWN ON REINFORCEMENT COVER SCHEDULE TABLE.

DESIGN PARAMETERS

THE FOLLOWING DESIGN PARAMETERS ARE ASSUMED FOR RETAINING WALL DESIGN. THE GEOTECHNICAL PARAMETERS SHALL BE CONFIRMED BY A GEOTECHNICAL ENGINEER:

DP1 LIVE LOAD VERTICAL SURCHARGE = 5 KPa MAXIMUM. NOT SUITABLE FOR VEHICLE LOAD WITHIN "H" METRE FROM WALL

DP2 ASSUMED FOUNDATION CHARACTERISTIC INTERNAL FRICTION ANGLE = 35°

DP3 ASSUMED CHARACTERISTIC INTERNAL FRICTION ANGLE OF INFILL MATERIAL & RETAINED SOIL = 30°

DP4 ASSUMED CHARACTERISTIC COHESION OF FOUNDATION = 3KPa.

DP5 DESIGN SHALL NOT APPLY TO SITES WITH INHERENT PROPERTIES FOR GLOBAL SLIP, SETTLEMENT, DRAINAGE & SIMILAR CONSIDERATIONS. FOR SUCH SITES A GEOTECHNICAL REPORT BY A QUALIFIED & EXPERIENCED ENGINEER SHOULD BE OBTAINED.

DP6 SAFETY & PROTECTION OF STRUCTURES SHALL BE VERIFIED BY QUALIFIED ENGINEER

DESIGN MUST BE VERIFIED BY A QUALIFIED ENGINEER FOR SITE SPECIFIC CONDITIONS

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.

CANTERBURY BANKSTOWN

CITY OF CANTERBURY BANKSTOWN

STANDARD RETAINING WALL CONCRETE BLOCK

RW-CB NOTES

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